



DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH
JAMSHORO



Class: B.S III

Course Title & NO. Cell & Molecular Biology.502

Date: 30.04.2019

1st Semester, 2019

Max. Marks: 50

Time: 02 Hours

Attempt any three questions. All questions carry equal marks.

- Q1. Define Molecular Biology. How it differs from Biochemistry? Enlist two scientists related to the field of Molecular Biology along with note on their contributions
- Q2. Which molecule is ranked as blue print of an organism? Support your answer with scientific experiment.
- Q3. Replication is anti- semi-conservative in origin. Prove or disapprove this statement with experimental findings.
- Q4. Place nucleotide, nucleoside and nucleosome in ascending order. Which term is least inclusive from them? Briefly explain the term which is monomer from them
- Q5. What are nitrogenous bases? How many types they have? Write a brief note on solo ringed nitrogenous bases.



**DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH
JAMSHORO**



Class: B.S II

(Major – Regular)

1st Semester, 2021

Course Title: Invertebrate Diversity (Course No. 410-411)

Max. Marks: 60

Date: 09.07.2021

Time: 02 Hours

Attempt any Four questions. All questions carry equal marks.

Q.NO:1. Give a detailed account of polymorphism in Coelenterates.

Q.NO:2. Distinguish between arthropod and annelids with diagram.

Q.NO:3. What is locomotion? Describe the amoeboid and flagella movement exhibited by protozoans with the help of diagram.

Q.NO:4. Define the circulatory system in molluscans.

Q.NO:5. Describe in detail the parasitic adaptation of helminthes.

Q.NO:6. Describe the general characters of phylum Echinodermata.

Q.NO:7. Define helminthes? Describe parasitic adaptation of liver fluke.

Q.NO:8. Describe the general characteristics of the phylum ctenophore and describe their classes up to order level with the examples.

Practical

Define the following terms.

1. Mesoglea

2. Parasite

3. Encystment

4. Contractile vacuole

5. Saprophytic

6. Spongocoel

7. Pellicle

8. Sangivorous

9. Acetabulum

10. Cephalopods

11. Jointed Appendages

12. Regeneration

13. Trilobite

14. Chelicerate

15. Cephalothorax

16. Crustaceans

**DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH**

Class: B.S III

Subject: Animal behavior (practical)

Time: 1 hr

1ST semester

Marks 70

Date: 06.05.2019

1. Define any ten terms of the followings:

1. Imprinting
- ✓ 2. Phono taxes
- ✓ 3. Primer pheromones
4. Stimulus
- ✓ 5. Piloting
6. Compass orientation
- ✓ 7. Ethology
- ✓ 8. Navigation
9. Trial and error learning
- ✓ 10. Social behavior
- ✓ 11. Geo taxes
- ✓ 12. Releaser pheromone
- ✓ 13. Communication
- ✓ 14. Chemo taxes

DEPARTMENT OF ZOOLOGY
BS III-1st SEMESTER 2019
CELL & MOLECULAR BIOLOGY (PRACTICAL)

Q 1) Draw the figures of the following Tissue.

- b) ADIPOSE TISSUE. b) NERVOUS TISSUE. c) BLOOD. d) SMOOTH MUSCLE TISSUE.

e) DNA

Q2) Write down the location and function of following Tissue

- c) SKELETAL MUSCLE TISSUE. b) SIMPLE SQUAMOUS EPITHELIUM. c) CARDIAC MUSCLE TISSUE.
d) SIMPLE CUBOIDAL TISSUE. e) SIMPLE COLUMNAR EPITHELIUM.

Q3) Define the following terms.

- b) LYSOSOMES b) VAULT c) MICRO TUBULES. d) INTERMEDIATE FILAMENTS.
e) MICROFILAMENTS.

Department of Zoology
University of Sindh,
Jamshoro

Max. Marks: 50
B.S.-III (1st Semester)
Time: 2 hrs.

Date of Examination: 03-05-2019

Subject: Physiology

Note: Attempt any four questions. All questions carry equal marks.

- Q.1 what is hypertension? Write down the method for measuring blood pressure?
- Q.2 what is energy metabolism and how it is supplied to muscles for contraction?
- Q.3 Describe the importance of different types of respiratory pigments in invertebrates and vertebrates?
- Q.4 what is diabetes mellitus? Write down the effect of insulin on glycaemia?
- Q.5 what is homeostasis? Discuss the process of urine formation by four steps in nephron?
- Q.6 what is osmoregulation? Discuss the structure of nephron, cortex and medulla in kidney?
- Q.7 what is the function of Pacemaker? How the heart muscles contracts and relaxes rhythmically?
- Q.8 what is myocardial infarction? Write down its causes, symptoms, controlled and preventions?
- Q.9 write short note on any two of the following:

1. Basic theme of Physiology
2. Cardiac cycle and Cardiac output.
3. Negative feedback system
4. Positive feedback system

**DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH**

Class: B.S III

Paper: Animal behavior

Date: 06. 05.2019

Semester 1st 2019

Marks: 50

Time: 90 Minutes

Attempt any two questions. All questions carry equal marks

1. Explain how migratory behavior of birds is genetic? Discuss in detail the benefits of migration.
- ✓ 2. Define the Pheromones? Discuss the role of different types of pheromones in animals.
3. What is a Social behavior? Why do we say that "the basis of all Social behaviors is Communication? Explain its types with suitable examples.
- ✓ 4. Define spatial migration? What type of cost animals do pay for migration?
5. Define the mechanism of orientation? Describe different types of long distance orientation.



DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH,
JAMSHORO



Class: BS-III
Subject: Psychology
Date: 08.05.2019

1st Semester, 2019
Max. Marks: 50
Time: 2 Hours

Attempt any four questions. All questions carry equal marks.

Q 1. What is Psychology? Differentiate between teacher and parent role in development of children psychology.

✓ Q 2. What is thinking? How it is developed? Discuss non-beneficial thinking with examples.

Q 3. How a teacher could be a role model for students? Give your own comments.

✓ Q 4. Define Savant syndrome. How Savants can be treated with social interactions?

✓ Q 5. Write down short notes on any five of the following:

- a. Intrapersonal intelligence
- c. Linguistic intelligence
- e. Problem solving

- b. Gender and intelligence
- d. Dreaming
- f. Interpersonal intelligence

Madam Nahed Shah



DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH
JAMSHORO



Class: BS-III

Subject: General Biochemistry

Date: 30.07.2021

Final

1st Semester, 2021

Max. Marks: 60

Time: 1-1/2 Hours

Attempt Any Three Questions:

Question-1: What are proteins? Discuss Hormonal and enzymatic types of Proteins.

Question-2: Define enzymes? Discuss mechanism action of the enzyme.

Question-3: Discuss classification & properties of Vitamins.

Question-4: Write down short notes on the following?

1. Denaturation of Protein
2. Glycoproteins
3. Non-essential amino acid
4. Tryptophan

**Department of Zoology
University of Sindh,
Jamshoro**

Date of Examination: 12-08-2021

Subject: Physiology

Note: Attempt any three questions. All questions carry equal marks.

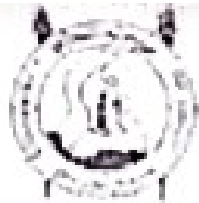
Max. Marks. 60
B.S -III (1st Semester)
Time: 2 hrs.

- Q.1 what is hypertension? Write down the method for measuring blood pressure?
- Q.2 what is energy metabolism? How energy is supplied for contraction of muscles?
- Q.3 what is diabetes mellitus? Write down the effects of insulin on glycaemia?
- Q.4 what is homeostasis? Discuss the process of urine formation by four steps in nephron?
- Q.5 what is osmoregulation? Discuss the structure of nephron, cortex and medulla in kidney?
- Q.6 what is the function of Pacemaker? How the heart muscles contracts and relaxes rhythmically?

Practical

Q.1 Define the following terms:

1. Basic theme of Physiology.
2. Cardiac cycle and Cardiac output.
3. Myocardial infarction.
4. Hemoglobin and Hemocyanin pigments.
5. Functions of white blood cells.
6. Thermoregulation and brown fats.



DEPARTMENT OF ZOOLOGY
UNIVERSITY OF SINDH,
JAMSHORO



Class: BS-I

Subject: Principles of Animal Life-I (ZOOL-310)

Date: 09.05.2019

Semester: 1st

Max. Marks: 50

Time: 2 Hours

Attempt any three questions. All questions carry equal marks.

- Q 1.** What is Zoology? Define any ten of its branches.
- Q 2.** Define ecosystem? Enlist various types of ecosystems along with brief description.
- Q 3.** What do you mean by Food chain? Explain different trophic levels along with examples.
- Q 4.** Which organisms play the role of producers in ecosystem? Point out various types of consumers along with detail of anyone of them.
- Q 5.** Differentiate between following:
- Pond ecosystem & River ecosystem
 - Rainforest & Tundra
 - Food chain and Food web

DEPARTMENT OF ZOOLOGY, UNIVERSITY OF SINDH, JAMSHORO

Animal Diversity Vertebrate 2nd Semester 2017 B.S II (Chemistry Supporting)

Time: 1 ½ Hours

Max Marks: 50

NOTE: ATTEMPT ANY TWO QUESTIONS. ALL CARRY EQUAL MARKS.

- ✓ 1. What are the defining characteristics of mammals? Discuss its social behavior?
2. What was the first class of vertebrates to live on land?
- ✓ 3. Discuss in detail important features of birds?

Practical

Define the following.

- ✓ 1. Hibernation
2. Viviparity
3. Pneumatic
- ✓ 4. Osteichthyes
5. Notochord
- ✓ 6. Furcula
- ✓ 7. Cold Blooded
8. Monotreme
- ✓ 9. Chordate
10. Cranium

wish bone.